

BEN AGRO

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Speedrunning to one million humanoid robots

Work Experience

Tesla AI - Senior Staff Robotics Machine Learning Engineer

📅 October 2026 - Present

- Working on AI for humanoid robots and large models for Optimus, in collaboration with xAI

Tesla AI - Staff Autopilot Machine Learning Engineer

📅 March 2025 - September 2026

- End-to-end driving policies, from pretraining to RL, under Phil Duan
- Currently focused on pretraining scaling; model, data, optimizer, evaluations, etc.
- Co-lead the development of FSD v14, resulting in a step-change improvement in safety and the first full unsupervised Tesla robotaxi
- Performance review ratings: 4/5 → 5/5 → 5/5

Waabi - Researcher II

📅 May 2022 - March 2025

- Developing novel perception and forecasting systems that enable self-driving trucks across perception, forecasting, behavior simulation, and sensor simulation.
- Rapid development from research ideas to improvements on real autonomous trucks.
- Published work through various papers; see MAD, DIO, UnO, ImplicitO, DeTra, QUaD in the "papers" section below.

Robotics Vision and Learning Lab - Undergraduate Researcher

📅 May 2021 - Sept 2021

- Enabled robots to operate in complex scenarios with learning-based task and motion-planning systems
- Built a simulation environment for development, extended **PDDLStream** with learned stream-scoring and a queue-based algorithm, and deployed the system to a Franka Panda robot
- Published our work in the paper "Learning to Search in Task and Motion Planning with Streams"

Autonomous Space Robotics Lab - Undergraduate Researcher

📅 May 2020 - Aug 2020

- Developed a self-supervised semantic LiDAR segmentation pipeline for autonomous navigation
- Published the paper "Self-Supervised Learning of LiDAR Segmentation for Autonomous Indoor Navigation"

Publications and Preprints

MAD: MEMORY-AUGMENTED DETECTION OF 3D OBJECTS

Ben Agro, Sergio Casas, Patrick Wang, Thomas Gilles, Raquel Urtasun

CVPR 2025

Top online non-ensemble method on the Waymo Open Detection Challenge

DIO: DECOMPOSABLE IMPLICIT 4D OCCUPANCY-FLOW WORLD MODEL

Christopher Diehl, Quinlan Sykora, **Ben Agro**, Thomas Gilles, Sergio Casas, Raquel Urtasun

CVPR 2025

DeTra: A UNIFIED MODEL FOR OBJECT DETECTION AND TRAJECTORY FORECASTING

Sergio Casas*, **Ben Agro***, Jiageng Mao*, Thomas Gilles, Alexander Cui, Thomas Li, Raquel Urtasun

ECCV 2024

UNO: UNSUPERVISED OCCUPANCY FIELDS FOR PERCEPTION AND FORECASTING

Ben Agro, Quin Sykora, Sergio Casas, Thomas Gilles, Raquel Urtasun

CVPR 2024, **Oral (top 0.7% of submitted papers)**

QUAD: QUERY-BASED INTERPRETABLE NEURAL MOTION PLANNING FOR AUTONOMOUS DRIVING

Sourav Biswas, Sergio Casas, Quin Sykora, **Ben Agro**, Abbas Sadat, Raquel Urtasun

ICRA 2024

IMPLICIT OCCUPANCY FLOW FIELDS FOR PERCEPTION AND PREDICTION IN SELF-DRIVING

Ben Agro, Quin Sykora, Sergio Casas, Raquel Urtasun

CVPR 2023, **Highlight**

TOWARD GLOBALLY OPTIMAL STATE ESTIMATION USING AUTOMATICALLY TIGHTENED SEMIDEFINITE RELAXATIONS

Frederike Dümbgen, Connor Holmes, **Ben Agro**, Tim Barfoot
Arxiv Preprint 2023

LEARNING TO SEARCH IN TASK AND MOTION PLANNING WITH STREAMS

Mohamed Khodeir*, **Ben Agro***, Florian Shkurti
IEEE Robotics and Automation Letters 2023

SELF-SUPERVISED LEARNING OF LIDAR SEGMENTATION FOR AUTONOMOUS INDOOR NAVIGATION

Hugues Thomas, **Ben Agro**, Mona Gridseth, Jian Zhang, Tim Barfoot
ICRA 2021

Education

- Master's in Computer Science, University of Toronto  Feb 2025 - Oct 2025
Supervised by [Raquel Urtasun](#)
- PhD studies in Computer Science (withdrawn), University of Toronto  Sept 2023 - Feb 2025
Supervised by [Raquel Urtasun](#)
- BASc in Engineering Science, University Of Toronto  Sept 2019 - May 2023
Majored in robotics, Cumulative Average = 97%, CGPA = 4.0

Awards

- John Black Aird Scholarship, W.S. Wilson Medal, Ontario Professional Engineers Foundation Gold Medal, Governor's General Silver Medal
 May 2024
Awarded to the student with the highest academic standing of any UofT undergraduate
- Centennial Senior Project Award  May 2024
Awarded for the best [undergraduate thesis project](#)
- University of Toronto Excellence Award  May 2020
To fund the research of exceptional UofT undergraduate students
- University of Toronto Scholar Award  Sept 2019
Recognition of UofT's outstanding students at admission
- Governor General's Bronze Medal  May 2019
Awarded for the highest academic standing in highschool
- AP National Scholar  May 2019
Graduated highschool with six AP (university equivalent) credits

Personal Projects

- BARFT: BUNDLE ADJUSTING NEURAL RADIANCE FIELDS WITH TEMPORAL REGULARIZATION  
A framework for training NeRFs with unknown (learned) camera poses.
- EXPLAINER OF "TRANSFORMERS AS STATISTICIANS" 
Distilling the key ideas behind "[Transformers as Statisticians: Provable In-Context Learning with In-Context Algorithm Selection](#)"
- ZERO-SHOT VIDEO RETRIEVAL WITH VISION LANGUAGE MODELS  
A zero-shot video retrieval system that leverages open-source vision-language models
- TOWARDS GLOBALLY OPTIMAL STEREO LOCALIZATION (UNDERGRAD THESIS)   
An investigation into how to make the problem stereo localization globally optimal, supervised by Tim Barfoot

"CAPTOR" THE AUTONOMOUS DRONE

A custom-built autonomous drone with reliable onboard SLAM and vision-only obstacle avoidance



GEOMETRY BOY

A version of the popular game Geometry Dash programmed for the original Gameboy hardware



News

Top U of T undergraduate Ben Agro is taking his passion for research into a direct-entry PhD

📅 June 12, 2023